

■ INTRODUCTION

The CE1001 is an integrated hall-effect sensor designed specifically to meet the requirements of low-power devices as an On/Off switch in Cellular Flip-Phones, which battery operating voltages of 1.8V-5.5V.

An onboard clock scheme is used to reduce the average operating current of IC CE1001 can be switched on with either the North or South pole of a magnet.

■ FEATURES

- 1.8V -5.5V battery operation
- High sensitivity and high stability of the magnetic switching points
- High resistance to mechanical stress
- Digital output signal
- Switching for both poles of a magnet

■ APPLICATIONS

- Mobile Phones (Flip Type, Slide Type etc.)
- Laptop PCs, Notebook PCs
- Digital Still And Video, Cameras
- Playthings, Portable games
- Electronic Dictionaries
- Home Appliances

■ BLOCK DIAGRAM

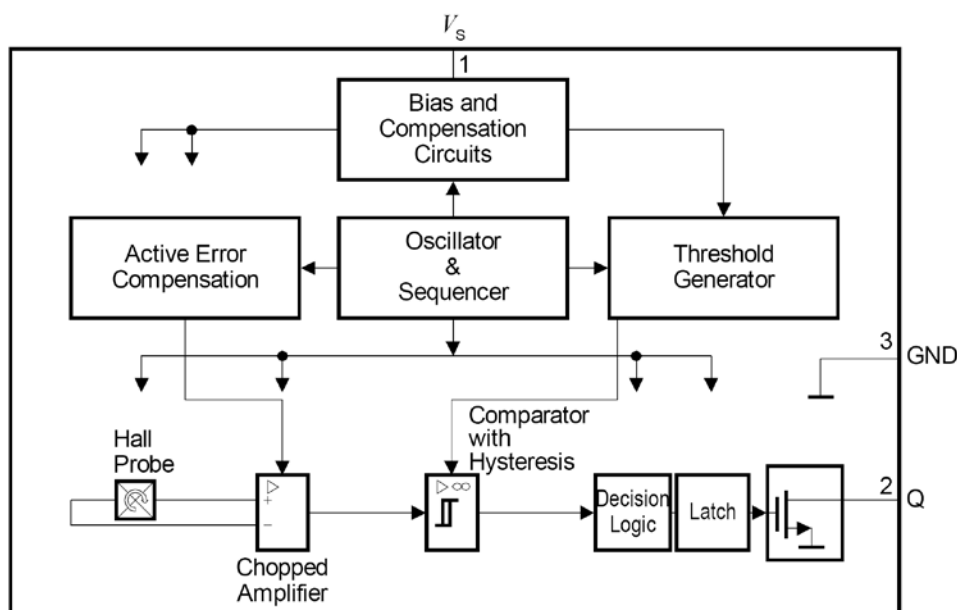
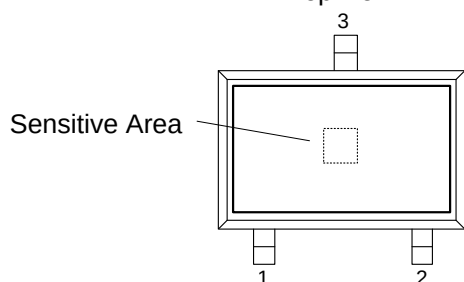


Figure1 Block Diagram

PIN CONFIGURATION

SOT23-3

Top view

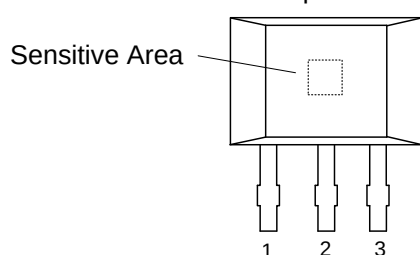


CE1001 Series (SOT23-3 PKG)

PIN NO.	PIN NAME	FUNCTION
1	V_S	Supply Voltage
2	Q	Open Drain Output
3	GND	Ground

TO-92S

Top view



CE1001TS Series (TO-92S PKG)

PIN NO.	PIN NAME	FUNCTION
1	V_S	Supply Voltage
2	GND	Ground
3	Q	Open Drain Output

ABSOLUTE MAXIMUM RATINGS⁽¹⁾

(Unless otherwise specified, $T_A=25^\circ\text{C}$)

PARAMETER	SYMBOL	MIN.	MAX.	UNITS
Supply Voltage ⁽²⁾	V_S	-0.3	5.5	V
Supply Current	I_S	- 1	2.5	mA
Output Voltage ⁽²⁾	V_Q	- 0.3	5.5	V
Output Current	I_Q	- 1	2	mA
Junction Temperature	T_j	-40	150	$^\circ\text{C}$
Storage Temperature	T_S	-40	150	$^\circ\text{C}$
Magnetic Flux Density	B	---	unlimited	mT
Thermal Resistance SOT23-3	$R_{th JA}$	---	35	K/W

(1) Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) All voltages are with respect to network ground terminal.

■ OPERATING RATINGS

PARAMETER	SYMBOL	MIN.	TYP.	MAX.	UNITS	NOTES
Supply Voltage	V_S	1.8	2.7	5.5	V	(1)
Output Voltage	V_Q	- 0.3	2.7	5.5	V	
Ambient Temperature	T_A	-40	25	85	°C	

1) A Ceramic Bypass Capacitor of 100nF at V_S to GND is highly recommended.

■ ELECTRICAL CHARACTERISTICS

($T_A=25^{\circ}\text{C}$, $V_S=2.7\text{V}$)

PARAMETER	SYMBOL	MIN.	TYP.	MAX.	UNITS	NOTES
Averaged Supply Current	I_{SAVG}		7		μA	
Averaged Supply Current during Operating Time	I_{SOPAVG}	0.5	2.0	3.5	mA	
Transient Peak Supply Current during Operating Time	I_{SOPT}	-	-	4.5	mA	
Supply Current during Standby Time	I_{SSTB}	1	1.9	8	μA	
Output Saturation Voltage	V_{QSAT}	-	0.13	0.4	V	$I_Q=1\text{mA}$
Output Leakage Current	I_{QLEAK}	-	0.01	1	μA	
Output Rise Time	t_r	-	0.5	1	μs	$R_L=2.7\text{K}\Omega$, $C_L=10\text{pF}$
Output Fall Time	t_f	-	0.1	1	μs	$R_L=2.7\text{K}\Omega$, $C_L=10\text{pF}$
Operating Time	t_{op}	25	100	160	μs	
Standby Time	t_{stb}		140	240	ms	
Duty Cycle	t_{op} / t_{stb}	-	0.071	-	%	
Start-up Time of IC	t_{stu}	-	12	20	μs	

■ MAGNETIC CHARACTERISTICS

(T_J=25°C, V_S=2.7V)

PARAMETER	MIN.	TYP.	MAX.	UNITS
B _{OPS}	1.4	2.5	5	mT
B _{OPN}	-5	-2.5	-1.4	mT
B _{RPS}	0.8	1.8	3.6	mT
B _{RPN}	-3.6	-1.8	-0.8	mT
B _{HYS}	0.2	0.8	1.6	mT

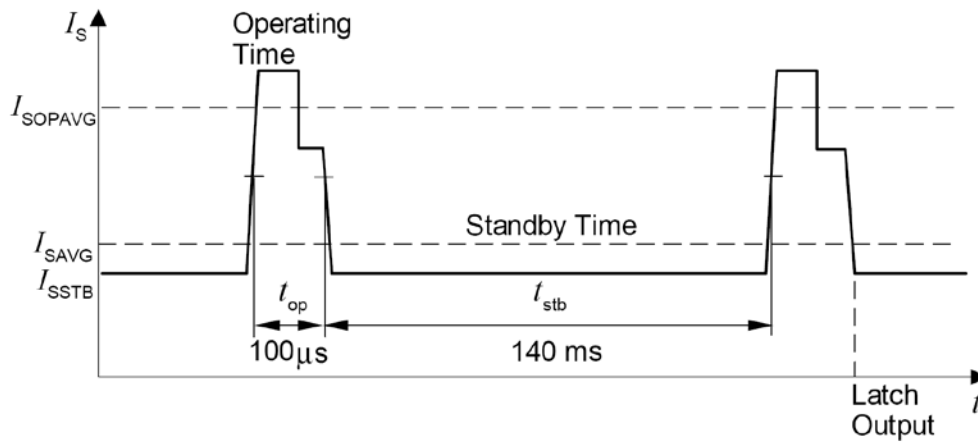


Figure2 Timing Diagram

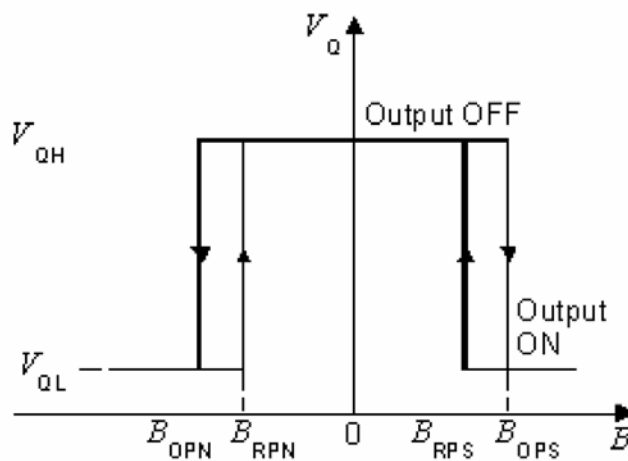
V_Q as function of the applied B-Field

Figure3 Output-Signal CE1001

■ TYPICAL APPLICATION CIRCUITS

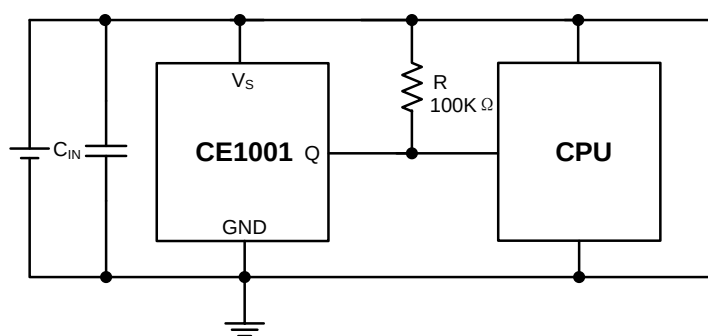
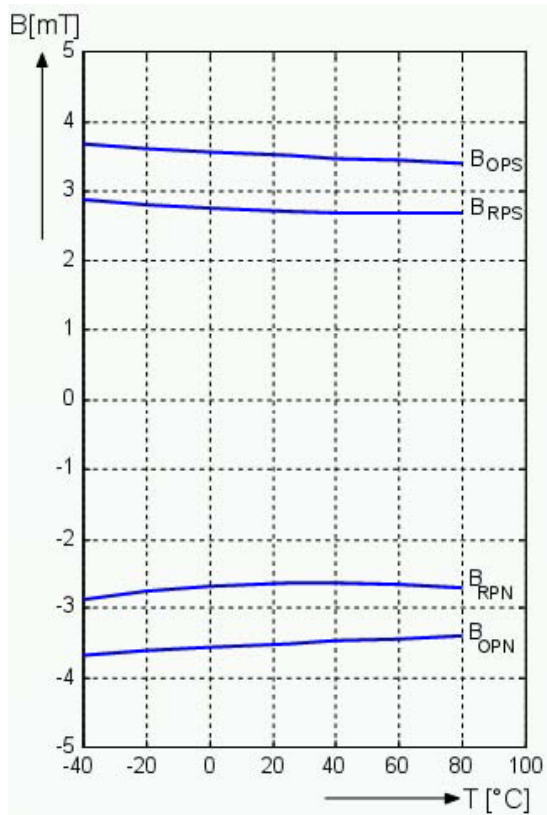


Figure4 Typical Application Circuit

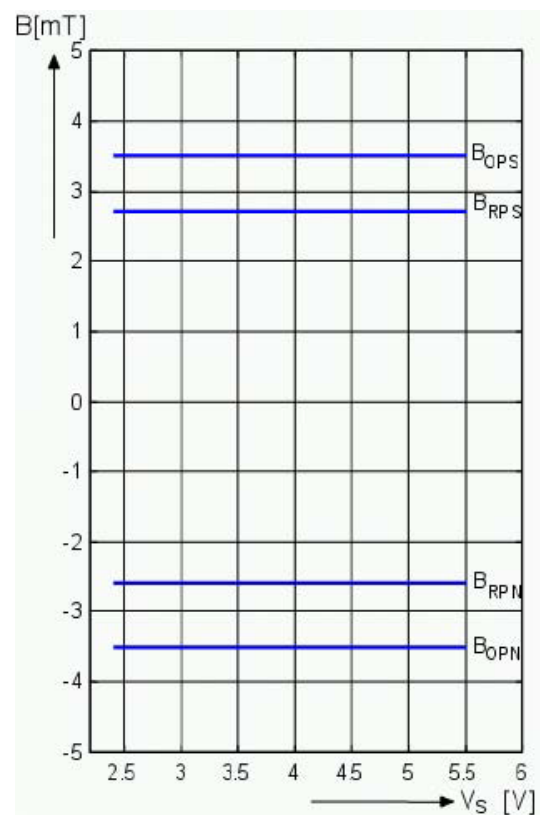
■ TYPICAL PERFORMANCE CHARACTERISTICS

All curves reflect typical values at the given parameters for T_A in $^{\circ}\text{C}$ and V_S in V.

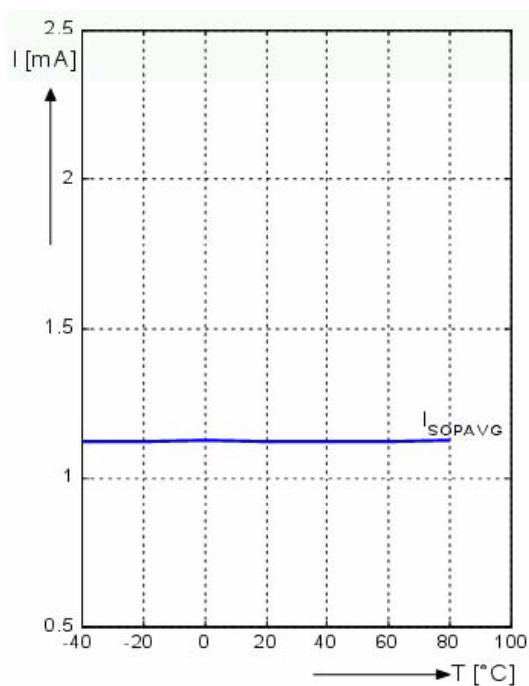
Magnetic Switching Points versus Temperature ($V_S=2.7\text{V}$)



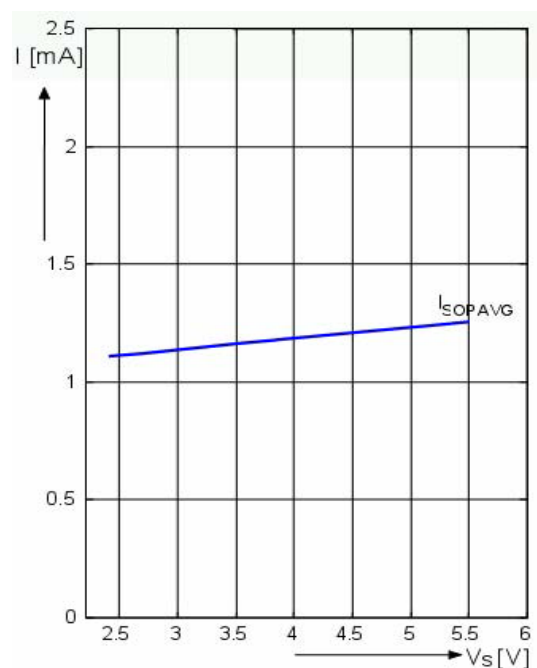
Magnetic Switching Points versus Supply Voltage ($T_A=20^{\circ}\text{C}$)



Supply current I_{SOPAVG} during Operating Time versus Temperature ($V_S=2.7\text{V}$)



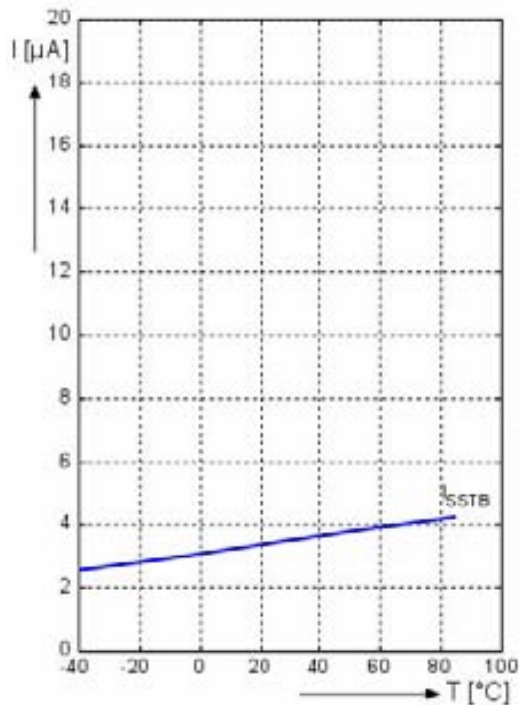
Supply current I_{SOPAVG} during Operating Time versus Supply Voltage ($T_A=20^{\circ}\text{C}$)



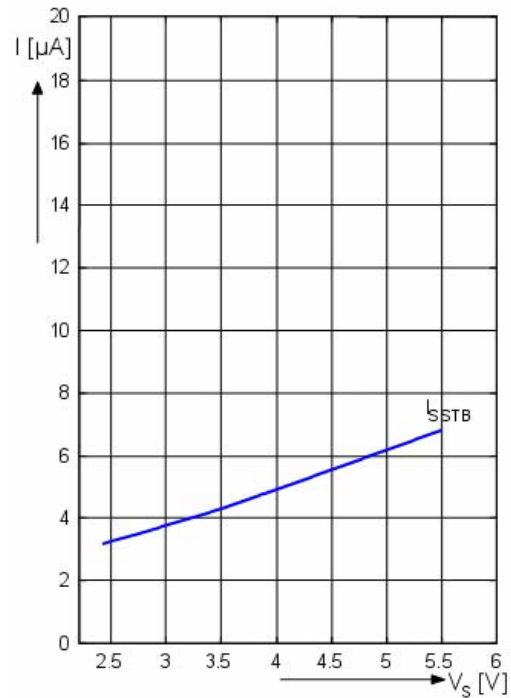
TYPICAL PERFORMANCE CHARACTERISTICS(CONTINUED)

All curves reflect typical values at the given parameters for T_A in $^{\circ}\text{C}$ and V_S in V.

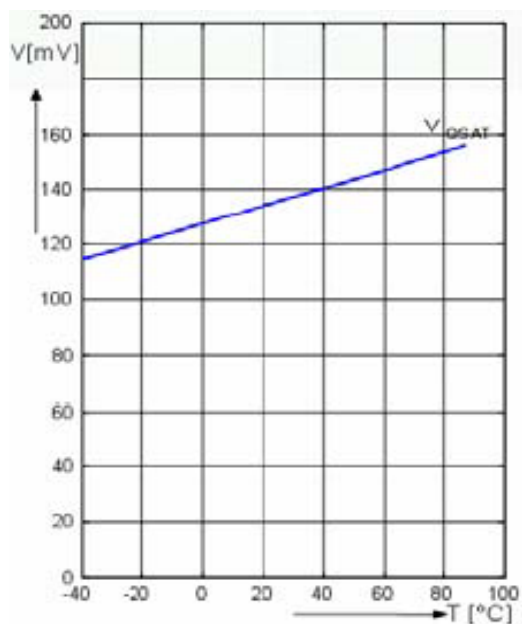
**Supply current I_{SSTB} during Standby
Time versus Temperature ($V_S=2.7\text{V}$)**



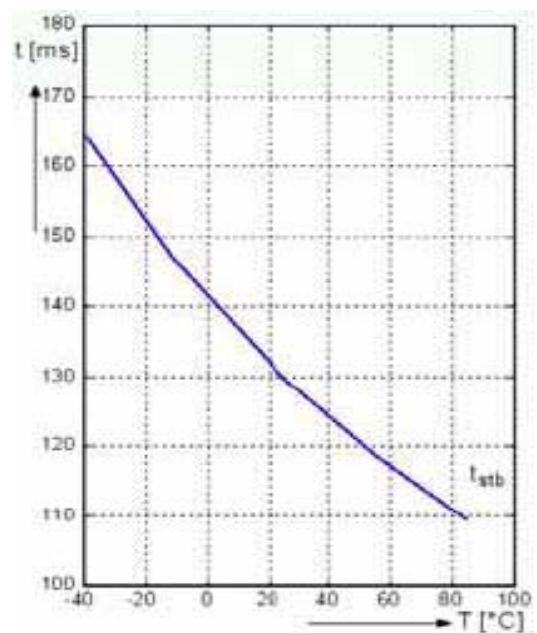
**Supply current I_{SSTB} during Standby
Time versus Supply Voltage ($T_A=20^{\circ}\text{C}$)**



**Output Saturation voltage V_{QSAT}
Temperature ($I_Q=1\text{mA}$)**

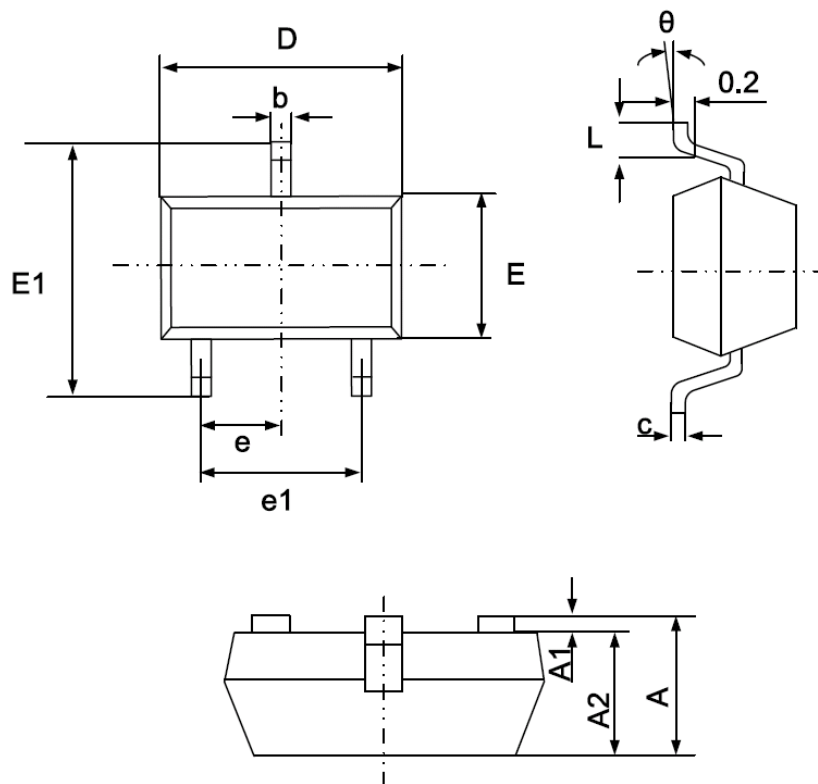


**Standby Time t_{stb} versus Temperature versus
($V_S=2.7\text{V}$)**



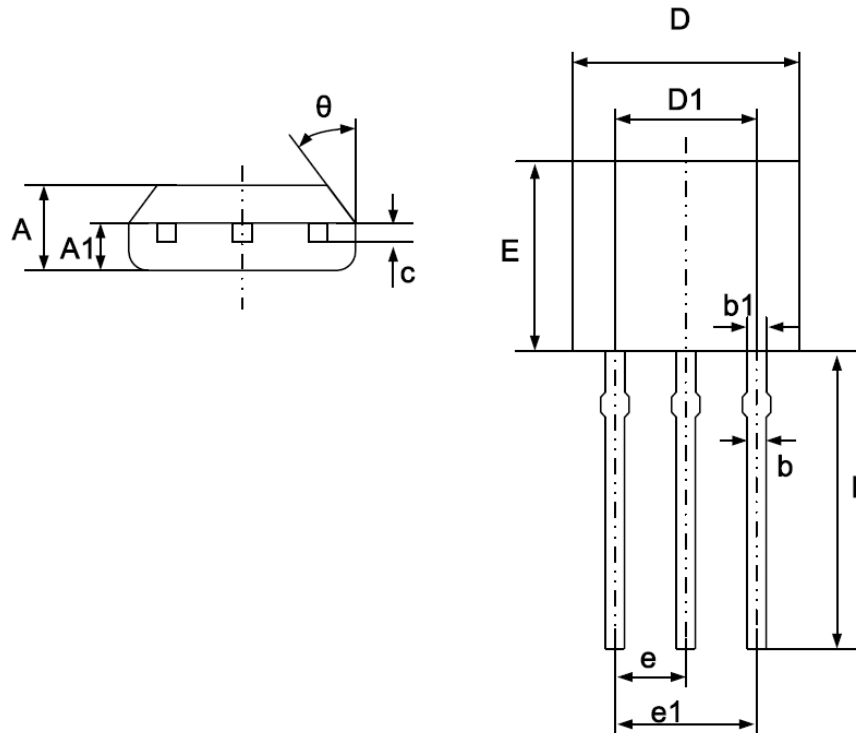
■ PACKAGE INFORMATION

● SOT23-3



Symbol	Dimension In Millimeters		Dimension In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

- TO-92S



Symbol	Dimension In Millimeters		Dimension In Inches	
	Min.	Max.	Min.	Max.
A	1.420	1.620	0.056	0.064
A1	0.660	0.860	0.026	0.034
b	0.350	0.480	0.014	0.019
b1	0.400	0.550	0.016	0.022
c	0.360	0.510	0.014	0.020
D	3.900	4.100	0.154	0.161
D1	2.280	2.680	0.090	0.106
E	3.050	3.250	0.120	0.128
e	1.270 TYP.		0.050 TYP.	
e1	2.440	2.640	0.096	0.104
L	15.100	15.500	0.594	0.610
L1	1.6 TYP.		0.063 TYP.	
θ	45° TYP.		45° TYP.	

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